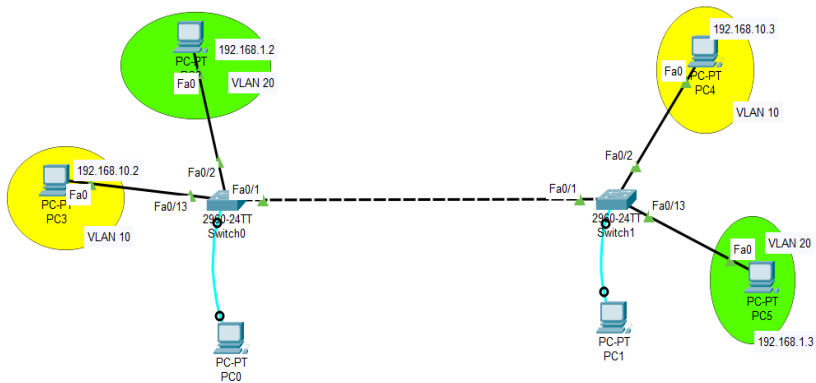


Construction of different VLANS and TRUNKING using cisco packet tracer



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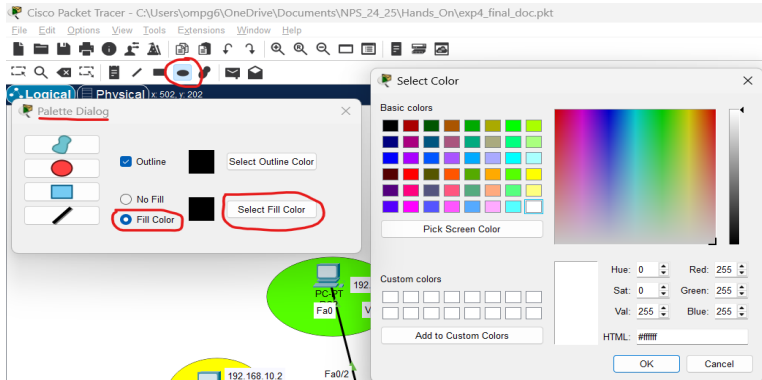
Console connection, Port display

- Console Connection: PC0 to Switch0 (left side switch)
 - Select console cable and connect to
 - **RS232** of PC0 ↔ **console** of Switch0
- Console Connection: PC1 to Switch1 (right side switch)
 - Select console cable and connect to
 - **RS232** of PC1 ↔ **console** of Switch1
- Connect FastEthernet port of PCs to FastEthernet ports Switch0 and Switch1 as shown in the figure
- **To display port labels**
 - Goto Options → Preferences → select **Always Show Port Labels in Logical Workspace**



Colour Display

Goto **Draw Ellipse** → Check the option **Fill Color** → **Select Fill Color**





Switch0 (left) Configuration

Open terminal of PC0

Click on PC0 → Desktop Tab → Terminal → Ok (Terminal Configuration)

- **Enter Global configuration mode**

- Switch>en
- Switch#config t

- **Create VLANs**

- Switch(config)#vlan 20
- Switch(config-vlan)#name green
- Switch(config-vlan)#exit
- Switch(config)#vlan 10
- Switch(config-vlan)#name yellow
- Switch(config-vlan)#exit



Switch0 (left) Configuration

● Assign Ports to VLANs

- Switch(config)#int f0/2
- Switch(config-if)#switchport mode access
- Switch(config-if)#switchport access vlan 20
- Switch(config-if)#exit
- Switch(config)#int f0/13
- Switch(config-if)#switchport mode access
- Switch(config-if)#switchport access vlan 10
- Switch(config-if)#exit

● Configure Trunk on the Interface

- Switch(config)#int f0/1
- Switch(config-if)#switchport mode trunk
- Switch(config-if)#switchport trunk allowed vlan 10,20
- Switch(config-if)#exit
- Switch(config)#exit



Check Switch0 (left) VLAN status

```
Switch#sh vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	yellow	active	Fa0/13
20	green	active	Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	



Check Switch0 (left) Interface Status

```
Switch#sh int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	trunk	auto	auto	10/100BaseTX
Fa0/2		connected	20	auto	auto	10/100BaseTX
Fa0/3		notconnect	1	auto	auto	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		connected	10	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX



Check Switch0 (left) Trunk Status

```
Switch#sh int trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	10,20

Port	Vlans allowed and active in management domain
Fa0/1	10,20

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	10,20



Switch1 (Right) Configuration

Open terminal of PC1

Click on PC1 → Desktop Tab → Terminal → Ok (Terminal Configuration)

- **Enter Global configuration mode**

- Switch>en
- Switch#config t

- **Create VLANs**

- Switch(config)#vlan 20
- Switch(config-vlan)#name green
- Switch(config-vlan)#exit
- Switch(config)#vlan 10
- Switch(config-vlan)#name yellow
- Switch(config-vlan)#exit



Switch1 (Right) Configuration

● Assign Ports to VLANs

- Switch(config)#int f0/2
- Switch(config-if)#switchport mode access
- Switch(config-if)#switchport access vlan 10
- Switch(config-if)#exit
- Switch(config)#int f0/13
- Switch(config-if)#switchport mode access
- Switch(config-if)#switchport access vlan 20
- Switch(config-if)#exit

● Configure Trunk on the Interface

- Switch(config)#int f0/1
- Switch(config-if)#switchport mode trunk
- Switch(config-if)#switchport trunk allowed vlan 10,20
- Switch(config-if)#exit
- Switch(config)#exit



Check Switch1 (Right) VLAN status

```
Switch#sh vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	yellow	active	Fa0/2
20	green	active	Fa0/13
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	



Check Switch1 (Right) Interface Status

```
Switch#sh int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	trunk	auto	auto	10/100BaseTX
Fa0/2		connected	10	auto	auto	10/100BaseTX
Fa0/3		notconnect	1	auto	auto	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		connected	20	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX



Check Switch1 (Right) Trunk Status

```
Switch#sh int trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	10,20

Port	Vlans allowed and active in management domain
Fa0/1	10,20

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	10,20



Assign IP Address

Click on PC → Desktop Tab → IP Configuration

- PC2 (Green-Left)
 - IPv4: 192.168.1.2; Subnet Mask:255.255.255.0; Gateway: 192.168.1.1
- PC5 (Green-Right)
 - IPv4: 192.168.1.3; Subnet Mask:255.255.255.0; Gateway: 192.168.1.1
- PC3 (Yellow-Left)
 - IPv4: 192.168.10.2; Sub Mask:255.255.255.0; Gateway: 192.168.10.1
- PC4 (Yellow-Right)
 - IPv4: 192.168.10.3; Sub Mask:255.255.255.0; Gateway: 192.168.10.1



Check Connection

Ping the PCs in the Circuit 2 to check connection

- PC2 and PC5 are connected
- PC3 and PC4 are connected



Conclusion

By following the steps outlined, we successfully constructed different VLANs and configured trunking on Cisco switches using Cisco Packet Tracer. The process involved:

- Creating VLANs for logical network segmentation.
- Configuring trunk ports to carry VLAN traffic between switches.
- Assigning VLANs to access ports to group devices logically